THE CANONICAL MAP DELTA[N] -> COSK_1 SK_1 DELTA[N] IS AN ISOMORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the factorization of the vertex maps eps^i_j, eps^i_{j'} through eps^i_{j,j'} : [1] -> [i] (source lines 2975-2980) is what reduces the simplicial-set condition to degree one; the converse construction is given explicitly (source lines 2962-2971); no verification is omitted.

Editorial note. The label lemma-simplex-cosk names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Simplicial sets and the 1-skeleton coskeleton adjunction. Same chapter, Sections on skeletons and coskeletons (sk_1, cosk_1). Statement lines 2938-2941: the canonical map Delta[n] -> cosk_1 sk_1 Delta[n]; the claim is checked by the defining universal property.
- Morphisms of simplicial sets as compatible families of maps. Statement and proof lines 2945-2952. A morphism f : U -> Delta[n] is a compatible family of maps f_u : [i] -> [n] with f_u phi = f_{U(phi)(u)}.

Lemma 0.1. *The canonical map $\Delta[n] \to cosk_1 sk_1 \Delta[n]$ is an isomorphism.*

Proof. Consider a simplicial set U and a morphism $f : U \to \Delta[n]$. This is a rule that associates to each $u \in U_i$ a map $f_u : [i] \to [n]$ in Δ. Furthermore, these maps should have the property that $f_u \circ \varphi = f_{U(\varphi)(u)}$ for any $\varphi : [j] \to [i]$. Denote $\epsilon^i_j : [0] \to [i]$ the map which maps 0 to j. Denote $F : U_0 \to [n]$ the map $u \mapsto f_u(0)$. Then we see that

$$f_u(j) = F(\epsilon^i_j(u))$$

for all $0 \leq j \leq i$ and $u \in U_i$. In particular, if we know the function F then we know the maps f_u for all $u \in U_i$ all i. Conversely, given a map $F : U_0 \to [n]$, we can set for any i, and any $u \in U_i$ and any $0 \leq j \leq i$

$$f_u(j) = F(\epsilon^i_j(u))$$

This does not in general define a morphism f of simplicial sets as above. Namely, the condition is that all the maps f_u are nondecreasing. This clearly is equivalent

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to the condition that $F(\epsilon^i_j(u)) \leq F(\epsilon^i_{j'}(u))$ whenever $0 \leq j \leq j' \leq i$ and $u \in U_i$. But in this case the morphisms

$$\epsilon^i_j, \epsilon^i_{j'} : [0] \to [i]$$

both factor through the map $\epsilon^i_{j,j'} : [1] \to [i]$ defined by the rules $0 \mapsto j$, $1 \mapsto j'$. In other words, it is enough to check the inequalities for $i = 1$ and $u \in X_1$. In other words, we have

$$\mathrm{Mor}(U, \Delta[n]) = \mathrm{Mor}(\mathrm{sk}_1 U, \mathrm{sk}_1 \Delta[n])$$

as desired. $\square$